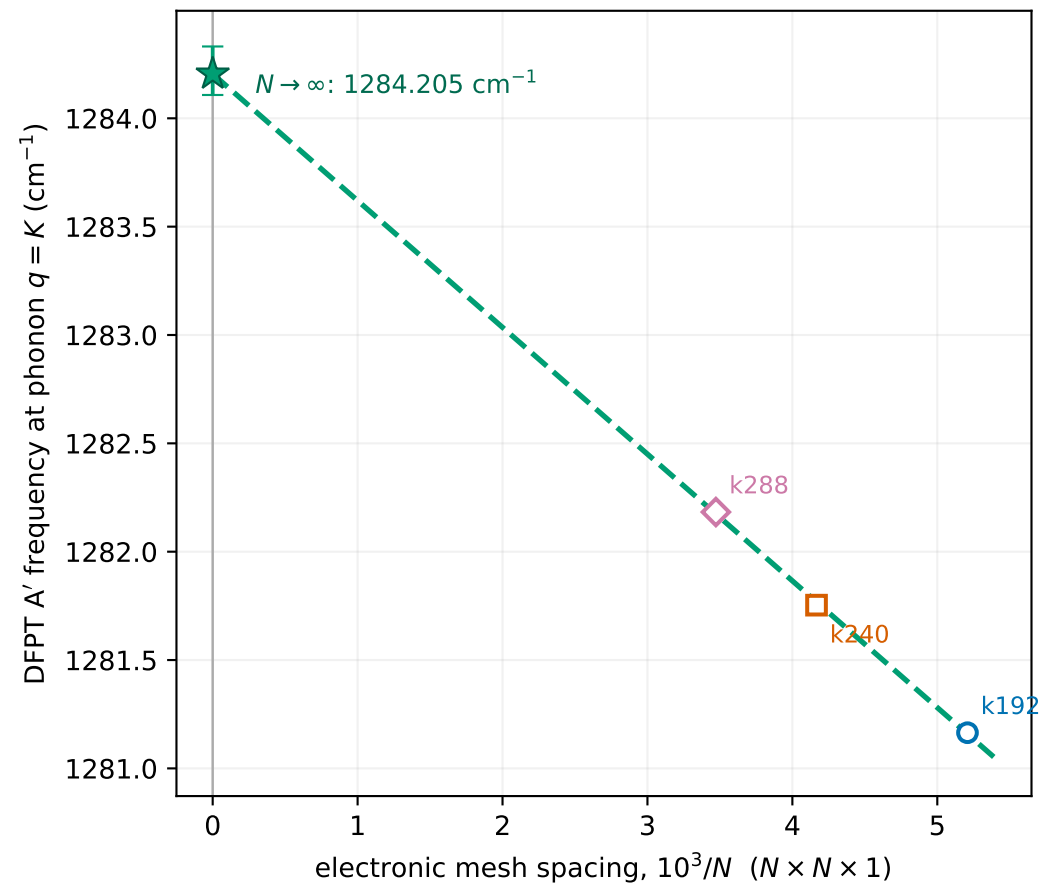
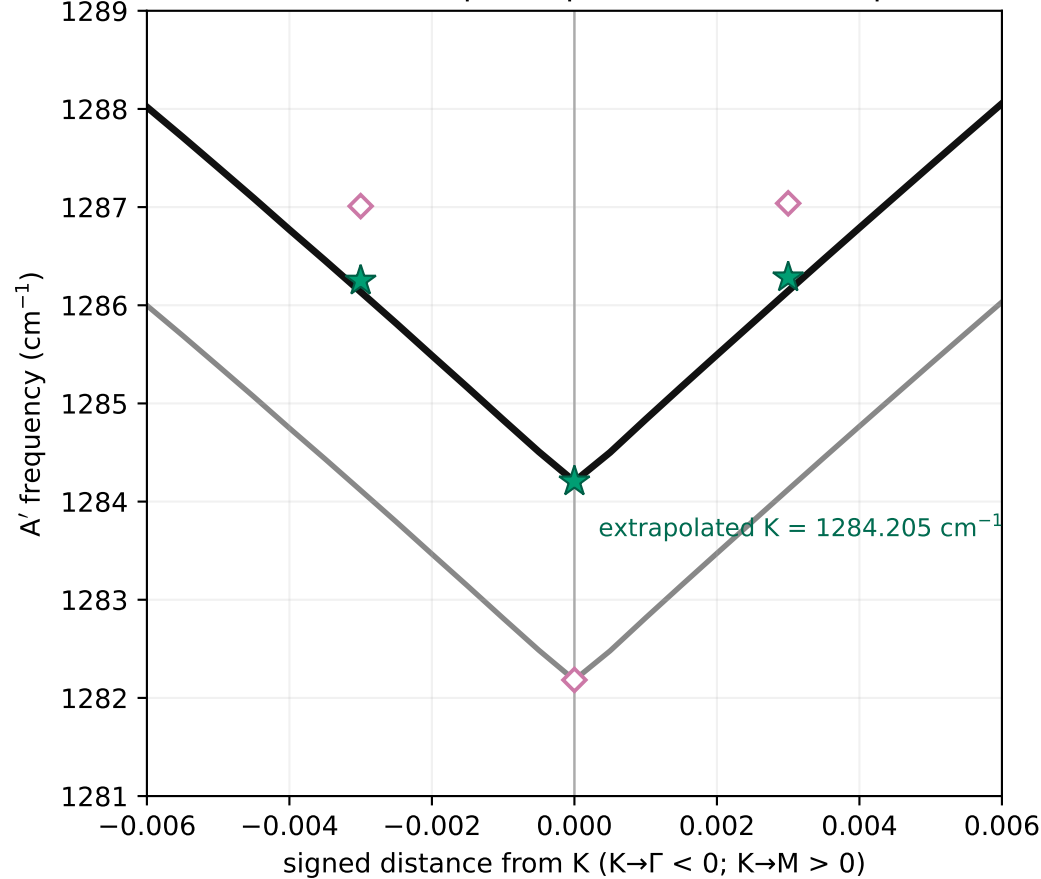


Graphene zero-smearing K-point frequency: electronic k-grid extrapolation

(a) Extrapolate the phonon $q = K$ frequency to $1/N = 0$



(b) Insert the fixed-q extrapolations into the spectrum



- smearing/degauss = 0 Ry
- linear fit: $\omega_K = \omega_K(\infty) + A/N$
 - fixed-q $N \rightarrow \infty$ extrapolation
 - MLIP + full EPC LR (k288 K anchor)
 - same model reanchored to DFPT K_∞
 - raw DFPT k288

Linear 3-grid K intercept: 1284.205 cm^{-1}
 Pairwise intercept range: $1284.108\text{--}1284.331 \text{ cm}^{-1}$
 Maximum fit residual: 0.015 cm^{-1}
 Quadratic-in- $1/N$ sensitivity: 1284.779 cm^{-1}

Reanchoring shift applied in panel (b):
 $+2.022 \text{ cm}^{-1}$; curve shape unchanged.

The pairwise range is a sensitivity diagnostic,
 not a statistical confidence interval.